



Module:	CSC211 (2026)	Assessment:	Term 2 Practical 3A
Lecturer:	Mr. C. K. Baker	Total:	Pass/Fail
Duration:	Mon 14h00 – 16h00		

Topic

Heaps, Problem-Solving, Debugging and Testing

Instructions

There are two (2) questions in this assignment. Submit a single compressed (.zip) file with all your code. Any sources used should be cited as comments in your code. Create and submit a file `Practical23A.java` that contains all your code.

Background

Assume a ternary heap stored in an `int[ ]` array starting at index 1. Each node has at most three children, and the heap satisfies either the min-heap or max-heap property.

Question 1

Write code for a method `void print(int heap[])`. It should print out the children for each node in the heap in the form "Node X -> A, B, C" where X is the parent node and A, B and C are its children, if they exist.

Question 2

Write code for a method `int[] validate(int heap[])`. It should return 1 if the heap is a valid min or max ternary heap, or -1 otherwise.

Hint

Your methods must work correctly for all `int[]` inputs, including very small arrays, large arrays, arrays containing duplicate or negative values, and cases where the last node has fewer than three children.